RENOVABITUR UT AQUILAE JUVENTUS TUA

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BioInformatics: Circulating microRNAs profiling in familial-hereditary breast cancer patient

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BACKGROUND

What are MicroRNAs?

- MicroRNAs (miRNAs) are small, non-coding RNA molecules, typically about 20-24 nucleotides long, that play a critical role in regulating gene expression.
- They function by binding to complementary sequences on messenger RNA (mRNA) molecules, leading to mRNA degradation or inhibition of translation.
- This post-transcriptional regulation allows miRNAs to influence various cellular processes, including development, differentiation, proliferation, and apoptosis.

OBJECTIVE

- The primary objective of this project is to profile circulating microRNAs (miRNAs) in familial-hereditary breast cancer (BC) patients. Specifically, the study seeks to perform a differential expression analysis of miRNAs collected from the plasma of breast cancer patients, with a particular focus on the presence or absence of germline pathogenic variants in the high-penetrance susceptibility genes BRCA1 and BRCA2.
- By identifying miRNAs it shows a significant difference in expression levels between the two groups, this analysis aims to provide insights into the molecular mechanisms underlying breast cancer and identify candidate biomarkers for diagnosis and prognosis.

EXPERIMENTAL SETTINGS

The project aims to identify circulating miRNAs in patients with hereditary-familial breast cancer by bioinformatics analysis of samples collected and sequenced in laboratory.

Study Population

- Description: Patients divided by BRCA status; controls selected to match patient age to avoid biases.
    - 34 BC patients recruited at UOC Oncologia Medica, S. Salvatore Hospital, L'Aquila
    - BRCA Positive: 16 patients.
    - BRCA Negative: 18 patients.
    - Controls: 7 healthy individuals, age-matched.

EXPERIMENTAL SETTINGS

Experimental protocol

1. RNA extraction and QC (mirVana Paris kit- Thermo)
2. Library preparation (QIAseq miRNA library kit – Qiagen)
3. Sequencing (Illumina NextSeq 500 – NovaSeq 600, 75/100 bp single-end, 10M reads/sample). The sequencing was conducted in three runs:
   - 4 samples (single-end 150 bp mode on NovaSeq 6000)
   - 20 samples (single-end 75 bp mode on NextSeq 500)
   - 19 samples (single-end 150 bp mode on NovaSeq 6000)
4. Bioinformatics analysis

BIOINFORMATICS ANALYSIS PIPELINE

- The pipeline designed for this project aims to be executable on standard PCs, such as laptops, by utilizing tools that require minimal computational resources.
- We also aimed to keep storage usage low, around 150GB, using tools capable of handling fastq.gz compressed files directly.
- This storage can be further reduced by deleting intermediate files.

BIOINFORMATICS ANALYSIS PIPELINE

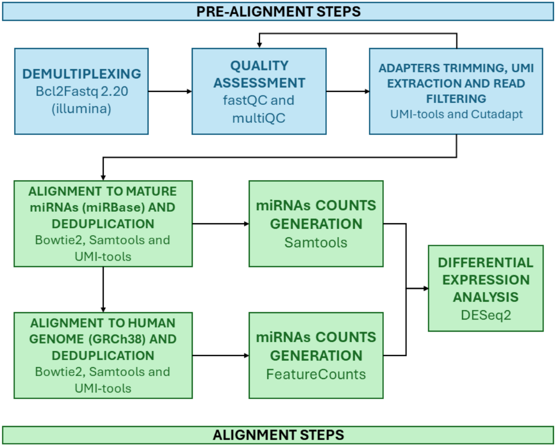

Figure 1: Bioinformatics Pipeline.

- When working with high-throughput sequencing technology like illumina, the first crucial step of a pipeline is demultiplexing.
- This process involves separating and identifying reads from different samples that were sequenced together in a single run.
- In the experiment we present, the demultiplexing step was already performed by the sequencing service using the Bcl2Fastq software provided by illumina.

We therefore focus on the next step, quality assessment of the provided fastq.gz samples.

- Firstly, we utilized FastQC, a reliable tool capable of handling fastq.gz files directly, to generate quality reports for each sample. Subsequently, we utilized MultiQC to compile and summarize these FastQC reports, producing three comprehensive reports, one for each sequencing run.
- From the reports, we observed the expected presence of the Illumina 3' adapter and the RT primer. Additionally, we noted a high duplication level of reads, which is anticipated due to the sample amplification required by Illumina preparation steps.
- Finally, we identified the presence of reads that were either too short or excessively long, with the latter exhibiting a low mean quality score, particularly in the samples from the third run.

To better understand this step, we briefly explain the structure of our reads [4] :

(3')TAGCAGCACGTAAATATTGGCGAACTGTAGGCACCATCAATT
CATCCGCCTGAAGATCGGAAGAGCACACGTCTGAACTCCAGTCAC(5')

- The first part of the read codes for the miRNA.
- The 3' adapter.
- The UMI tag, a sequence of fixed length of 12nt.
- The RT primer.

Trimming and Filtering

Issues identified during the quality assessment were addressed in this step:

- UMI-tools: Extracts the Unique Molecular Identifier (UMI) tag and trims adapters.
- Cutadapt: Trims reads to retain those with lengths between 15 and 30 nucleotides, suitable for miRNAs.

A subsequent quality assessment confirmed improved overall quality scores, allowing all samples to proceed to alignment.

Quality Check

The figure below presents the MultiQC report for the Run 3 samples before trimming. The Run 3 samples before trimming exhibit inconsistently in the quality across the entire length of the reads. There are major issues with sequencing quality.

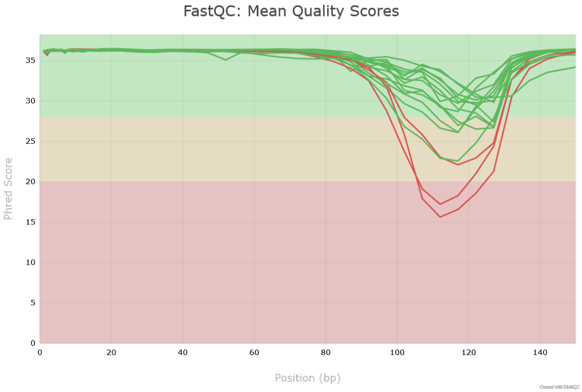

Figure 2: Quality check of run 3 before trimming.

Quality Check

The figure shows Run 3 samples after trimming and it demonstrates a successful enhancement in the quality of the sequencing data, with high and consistent quality scores across all bases. This indicates that the trimming process was effective in removing the low-quality regions observed in the initial data.

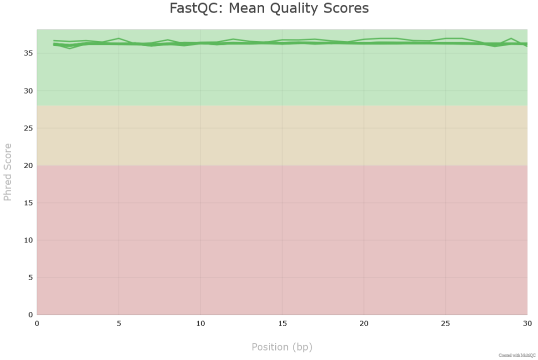

Figure 3: Quality check of run 3 after trimming.

Alignment

- Bowtie2: Chosen for its efficiency in aligning short sequences and memory-efficient execution on standard PCs. It provides pre-built indexes for the human genome (GRCh38).
- miRBase Alignment: Aligns samples to human mature miRNA entries from miRBase with stringent settings (no mismatches).
- GRCh38 Alignment: Aligns reads to the human genome with less stringent settings (one mismatch).
- The miRBase website provides a FASTA file containing mature miRNAs from multiple species, so the initial step was to select only human miRNAs. Next, since the file contains RNA sequences, we converted Uracil to Thymine in those sequences to make them suitable for alignment with the cDNA sequences from our samples.[3][1]

Alignment

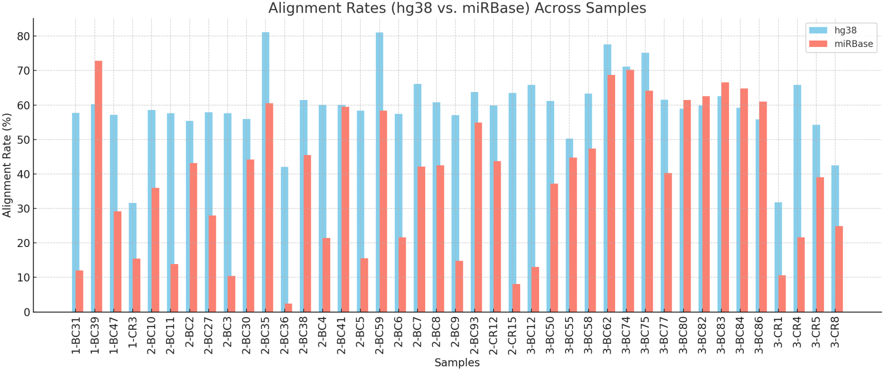

Figure 4: Alignment Rates.

Alignment

- The alignment rates to miRBase varied, with some samples exceeding 60% and others falling below this threshold.
- In contrast, the alignment to GRCh38 provided more consistent results, with an overall alignment rate of 60%.
- We hypothesize that the variability in the miRBase alignment results may be due to signs of degradation of miRNAs in plasma, although circulating miRNAs are known to be very stable.[2]
- Moreover, our samples can possibly contain other small ncRNAs (non-coding RNAs) that aren't mapped by miRBase.[5]

Counting and Differential Expression Analysis

- Before counting the results of the alignments, we deduplicated the sequences using UMI-tools to distinguish between true biological duplicates and PCR duplicates, by their UMI tags.
- In the counting step, we used Samtools to count the sequences mapped to miRBase and FeatureCounts to count the sequences mapped to the human genome, utilizing the annotation file provided by miRBase.
- We then generated a count matrix by summing the counts from both alignment steps and proceeded with the differential expression analysis using the R package DESeq2.

Counting and Differential Expression Analysis

- DESeq2 is a software package designed for analysing count data from RNA sequencing (RNA-seq) and other high-throughput sequencing assays. It employs statistical methods to identify differential expression between various conditions or treatments.
- By modeling counts with a negative binomial distribution, DESeq2 offers robust methods for normalization, differential expression testing, and variance estimation, making it a powerful tool for detecting significant changes in gene expression.

Counting and Differential Expression Analysis

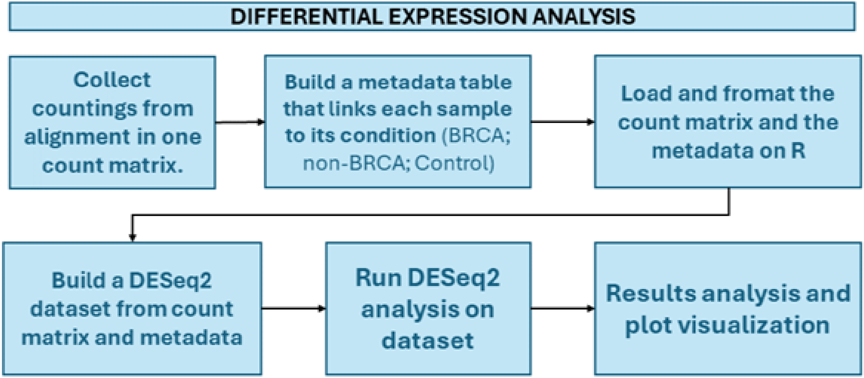

Figure 5: DESeq2 flow chart.

Results

The differential expression analysis of circulating microRNAs (miRNAs) in familial-hereditary breast cancer (BC) patients provided insightful findings. Here is an extended and detailed discussion of the results:

DESeq2 Differential Expression Analysis Results

	baseMean	log2FoldChange	lfcSE	stat	pvalue	padj
hsa-let-7a-3p	2.16945	-0.167528	1.594234	-0.105084	0.916309	0.941035
hsa-let-7a-5p	2486.83	2.362858	0.463218	5.100968	3.37921e-07	7.28783e-0
hsa-let-7b-3p	1.55025	-0.206711	0.677887	-0.304934	0.760416	0.832621
hsa-let-7b-5p	3361.01	0.802004	0.306471	2.6169	0.00887323	0.0630877
hsa-let-7c-3p	0.0125792	-1.794664	3.240328	-0.553853	0.57968	
hsa-miR-9985	1.328118	-0.288728	1.163658	-0.248121	0.8040411	
hsa-miR-99a-3p	0.230518	-0.811573	3.071845	-0.264197	0.7916279	
hsa-miR-99a-5p	30.929396	-1.31671	0.610684	-2.156123	0.0310741	
hsa-miR-99b-3p	1.43113	0.28252	0.712371	0.396591	0.6916688	
hsa-miR-99b-5p	25.433017	0.298224	0.674571	0.442094	0.6584214	

Figure 6: DESeq2 Differential Expression Analysis Results.

Summary of the Result of obtain from the DESeq Dataset

The Following results were obtained after following the flow chart of the DESeq2. A DESeq Dataset was obtain which has a dimension of 1622 rows and 41 columns

- log2FoldChange: Indicates the logarithmic change in expression levels. Positive values suggest upregulation, negative values indicate downregulation.
- lfcSE: The standard error of the log2 fold change, measuring variability in the fold change estimate.
- stat: The test statistic, often from the Wald test, showing how many standard deviations the log2 fold change is from zero.
- pvalue: The probability of observing the data under the null hypothesis. Lower values indicate stronger evidence against the null.
- padj: The adjusted p-value for multiple testing, controlling the false discovery rate to reduce type I errors.

Summary of the Result of obtain from the DESeq Dataset

```
out of 1622 with nonzero total read count
adjusted p-value < 0.1
LFC > 0 (up)       : 21, 1.3%
LFC < 0 (down)     : 115, 7.1%
outliers [1]       : 0, 0%
low counts [2]     : 975, 60%
(mean count < 1)
[1] see 'cooksCutoff' argument of ?results
[2] see 'independentFiltering' argument of ?results
```

Figure 7: Summary of the results.

Summary of the Result of obtain after running DESeq

Total miRNAs with nonzero total read count: 1622 Out of all the miRNAs analysed, 1622 have nonzero counts across the samples.

- Adjusted p-value < 0.1: This criterion is used to identify significant differential expression.
- LFC > 0 (up): a positive log fold change shows upregulation of miRNAs.
- LFC < 0 (down): a negative log fold change shows downregulatetion of miRNAs.
- Outliers [1]: the 'cooksCutoff' argument shows the presence of outliers. Outliers could influence the results if present.

Summary of the Result of obtain from the DESeq Dataset

- Low counts [2]: miRNAs that have mean counts less than 1 across samples, which are filtered out based on the "independent Filtering" argument.

  Low count miRNAs may not provide reliable statistical power and are often excluded from further analysis.

Visualization of the MA plot of the result

- The Plot below shows the MA plot of the log fold change and the mean of normalized counts of the DESeq Dataset most miRNAs are clustered around a log fold change of 0, indicating no significant difference in expression.
- Gray dots represent miRNAs with non-significant changes in expression, while blue dots represent miRNAs with significant differential expression (adjusted p-value < 0.1).
- The spread of points increases with higher mean normalized counts, which is typical due to greater variability in expression levels for more abundant miRNAs.

Visualization of the MA plot of the result

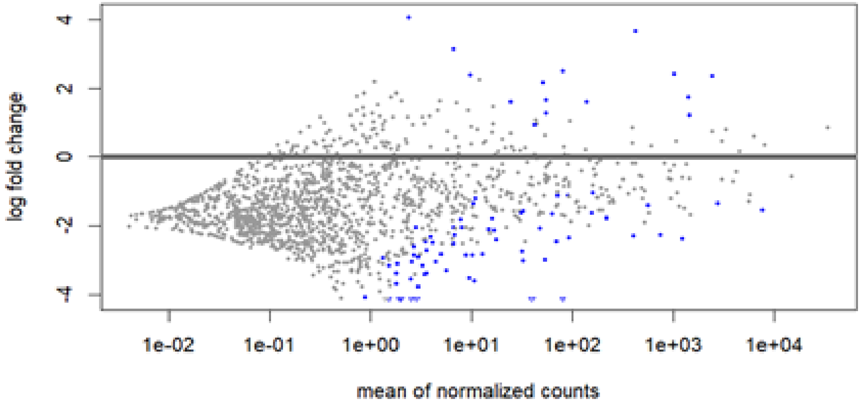

Figure 8: MA plot of DESeq2 Differential Expression Analysis Results.

BRCA vs. Controls

Column1	baseMean	log2FoldChange	lfcSE	stat	pvalue	padj
hsa-let-7a-3p	2.16944519	-1.546123339	2.116913054	-0.730366954	0.465165912	0.726917749
hsa-let-7a-5p	2486.830281	-0.764900595	0.634368419	-1.205767142	0.227907288	0.726917749
hsa-let-7b-3p	1.550245874	-0.957279226	0.926528896	-1.033188744	0.30151558	0.726917749
hsa-let-7b-5p	3361.010589	-0.585384584	0.420022459	-1.393698292	0.163408732	0.726917749
hsa-let-7c-3p	0.012579242	-3.253875326	4.2482533	-0.765932513	0.443716459	0.726917749
hsa-let-7c-5p	141.1444314	-0.722127843	0.486543929	-1.484198652	0.137756227	0.726917749
hsa-let-7d-3p	16.49755457	0.254751408	1.055189977	0.241427055	0.809224146	0.905839589
hsa-let-7d-5p	403.0887639	-0.002145704	0.530795649	-0.004042429	0.996774617	1
hsa-let-7e-3p	0.074496946	-3.123390208	4.247943618	-0.735271107	0.462174401	0.726917749
hsa-let-7e-5p	51.64326535	-0.404219876	0.645521869	-0.626190832	0.53118978	0.726917749
hsa-let-7f-1-3p	0.231685345	-2.865823064	3.585516202	-0.799277678	0.424129419	0.726917749
hsa-let-7f-2-3p	0.715970445	-2.072377744	2.662276943	-0.778423052	0.436319653	0.726917749
hsa-let-7f-5p	1021.977136	-0.240206989	0.62685979	-0.383190935	0.701578199	0.817499884
hsa-let-7g-3p	0.768717547	-1.43483311	1.968491113	-0.728899968	0.466062854	0.726917749
hsa-let-7g-5p	1441.470232	-1.140000248	0.70655289	-1.613467674	0.106642941	0.726917749
hsa-let-7i-3p	8.725517966	-0.00639573	1.481379477	-0.004317415	0.996555212	1
hsa-let-7i-5p	6348.967287	0.274147355	0.38329756	0.715233759	0.474464612	0.726917749
hsa-miR-1-3p	7.898716353	0.354343374	1.115471453	0.317662431	0.750741012	0.85693309
hsa-miR-100-5p	6.601794852	1.433180938	0.862862019	1.6609619	0.096721099	0.726917749
hsa-miR-101-2-5p	0.840473684	-1.984990591	1.890688067	-1.049877357	0.293774503	0.726917749
hsa-miR-101-3p	3957.299432	1.339065553	1.342319906	0.997575576	0.318485209	0.726917749
hsa-miR-101-5p	3.129411899	-0.88472904	1.675115336	-0.528160074	0.597388238	0.744209558
hsa-miR-10226	0.00825442	-3.253875903	4.248253302	-0.765932649	0.443716378	0.726917749
hsa-miR-10392-3p	0.057810657	-3.609436288	4.247119987	-0.849855031	0.39540569	0.726917749

Figure 9: Results obtained from BRCA vs. Controls.

BRCA vs. Controls

The result in the fig above shows the output result of Comparison of BRCA vs Controls.

The summary of the result shows only 2 miRNA's with significant differential expression (one upregulated and one downregulated) out of 1622 total genes analysed, with no outliers or genes with low counts impacting the results.

```
out of 1622 with nonzero total read count
adjusted p-value < 0.1
LFC > 0 (up)       : 1, 0.062%
LFC < 0 (down)     : 1, 0.062%
outliers [1]       : 0, 0%
low counts [2]     : 0, 0%
(mean count < 0)
[1] see 'cooksCutoff' argument of ?results
[2] see 'independentFiltering' argument of ?results
```

Figure 10: Summary of results obtained from BRCA vs. Controls.

The MA-plot below shows the visualization of Comparison of BRCA vs Controls, the majority of the miRNA's have no significant changes in expression (clustered around the horizontal line). However, a few genes are significantly differentially expressed (highlighted points), with one gene upregulated and one gene downregulated.

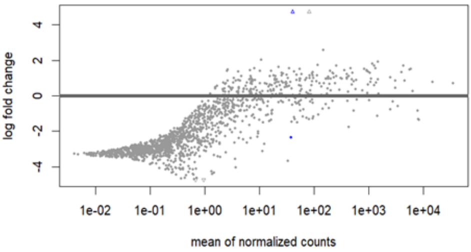

Figure 11: The MA plot of the result.

BRCA vs. non-BRCA

The analysis revealed miRNAs that are differentially expressed between BRCA and non-BRCA cases.

Column1	baseMean	log2FoldChange	lfcSE	stat	pvalue	padj
hsa-let-7a-3p	2.16944519	0.167528338	1.594233916	0.105083913	0.916309224	0.941035028708726
hsa-let-7a-5p	2486.830281	-2.362857635	0.463217505	-5.100967925	3.37921E-07	7.28782597274004e-05
hsa-let-7b-3p	1.550245874	0.206710986	0.677886744	0.304934397	0.760416117	0.832621336074657
hsa-let-7b-5p	3361.010589	-0.802004375	0.306471137	-2.616900188	0.008873228	0.0630876783577128
hsa-let-7c-3p	0.012579242	1.794663757	3.240327627	0.553852562	0.579679753	NA
hsa-let-7c-5p	141.1444314	-1.587646318	0.349429883	-4.543533329	5.53191E-06	0.000357914279293196
hsa-let-7d-3p	16.49755457	0.664474352	0.771701473	0.861051035	0.389209936	0.564616207240371
hsa-let-7d-5p	403.0887639	-0.84535469	0.384388698	-2.19921838	0.027862398	0.12347240758461
hsa-let-7e-3p	0.074496946	1.122891649	3.2336278	0.347254452	0.728400166	NA
hsa-let-7e-5p	51.64326535	-2.144980995	0.45811566	-4.682182213	2.83837E-06	0.00025014822421774
hsa-let-7f-1-3p	0.231685345	0.959781859	2.728978925	0.35169999	0.725063268	NA
hsa-let-7f-2-3p	0.715970445	2.196159756	2.034224744	1.079605271	0.280317993	0.462667708035898
hsa-let-7f-5p	1021.977136	-2.401923725	0.456648291	-5.259898641	1.44135E-07	7.28782597274004e-05
hsa-let-7g-3p	0.768717547	1.950966899	1.498835771	1.301654882	0.193034389	0.375054804608496
hsa-let-7g-5p	1441.470232	-1.725536527	0.517498971	-3.334376729	0.000854908	0.0122916749820861
hsa-let-7i-3p	8.725517966	1.488857853	1.102631238	1.35027723	0.176927073	0.351140541446603
hsa-let-7i-5p	6348.967287	-0.604010183	0.27973137	-2.159250796	0.030830715	0.127461752145125
hsa-miR-1-3p	7.898716353	1.373160167	0.813264194	1.688455212	0.09132388	0.243154526982832
hsa-miR-100-5p	6.601794852	0.548032536	0.579436442	0.945802674	0.344249242	0.527794453288652
hsa-miR-101-2-5p	0.840473684	1.816539198	1.446583325	1.2557446	0.209208583	0.391207957788977
hsa-miR-101-3p	3957.299432	0.696276614	1.003392955	0.693922168	0.487731021	0.637640548603598
hsa-miR-101-5p	3.129411899	1.017303818	1.253886603	0.81132043	0.417181681	0.580465693171567
hsa-miR-10226	0.00825442	1.87717441	3.240771311	0.579236925	0.562429318	NA
hsa-miR-10392-3p	0.057810657	1.548544713	3.239670277	0.477994543	0.632654089	NA
hsa-miR-10393-3p	0.046755813	2.472958275	3.240423933	0.763158872	0.445368653	NA

Figure 12: BRCA vs. non-BRCA Analysis Results.

BRCA vs. non-BRCA

- The output below shows the summary of the analysis of the above result of the BRCA and non-BRCA.
- A small proportion of miRNAs with significant differential expression between BRCA and non-BRCA cases, with more miRNAs upregulated in BRCA.
- The absence of outliers indicates robust data quality. A large percentage of miRNAs were filtered out due to low counts, focusing the analysis on reliably expressed miRNAs.

Conclusion

- These findings can be used for further investigation into the roles of these miRNAs in BRCA and non-BRCA conditions, potentially aiding in the identification of biomarkers or therapeutic targets.

BRCA vs. non-BRCA

```
out of 1622 with nonzero total read count
adjusted p-value < 0.1
LFC > 0 (up)       : 115, 7.1%
LFC < 0 (down)     : 21, 1.3%
outliers [1]       : 0, 0%
low counts [2]     : 975, 60%
(mean count < 1)
[1] see 'cooksCutoff' argument of ?results
[2] see 'independentFiltering' argument of ?results
```

Figure 13: Summary of Results.

BRCA vs. non-BRCA

Interpretation of the MA Plot

- The MA plot below visualizes the log fold change versus the mean of normalized counts for miRNAs between BRCA and non-BRCA cases.
- Most miRNAs show no significant differential expression, as indicated by gray dots clustering around the 0, log fold change line.
- Significant miRNAs (blue dots) are found across various expression levels, with higher variability in more abundant miRNAs.
- This plot is essential for identifying miRNAs with substantial and statistically significant expression changes for further investigation.

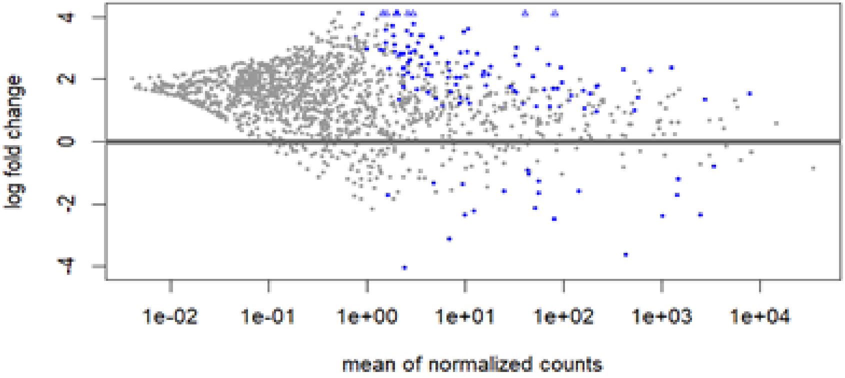

Figure 14: Interpretation of the MA Plot.

Non-BRCA vs. Controls

Column1	baseMean	log2FoldChange	lfcSE	stat	pvalue	padj
hsa-let-7a-3p	2.16944519	-1.713651677	2.068744941	-0.828353289	0.407470452	0.574859940498067
hsa-let-7a-5p	2486.830281	1.597957039	0.624141945	2.560246194	0.010459803	0.0765614481734251
hsa-let-7b-3p	1.550245874	-1.163990211	0.876621308	-1.327814189	0.184239498	0.36201974582996
hsa-let-7b-5p	3361.010589	0.216619791	0.413320692	0.52409617	0.600211653	0.734741850631882
hsa-let-7c-3p	0.012579242	-5.048539083	4.17531009	-1.209141112	0.226608637	NA
hsa-let-7c-5p	141.1444314	0.865518475	0.476474935	1.816503685	0.069293143	0.209353751875656
hsa-let-7d-3p	16.49755457	-0.409722944	1.038328617	-0.394598528	0.6931392	0.808093320449795
hsa-let-7d-5p	403.0887639	0.843208987	0.522084775	1.615080592	0.106293243	0.264800711494126
hsa-let-7e-3p	0.074496946	-4.246281857	4.170428044	-1.018188496	0.308588386	NA
hsa-let-7e-5p	51.64326535	1.74076112	0.628486063	2.769768849	0.005609609	0.0538219216802245
hsa-let-7f-1-3p	0.231685345	-3.825604923	3.515499959	-1.088210772	0.276502071	NA
hsa-let-7f-2-3p	0.715970445	-4.2685375	2.611247159	-1.634673871	0.102117433	0.257440645537309
hsa-let-7f-5p	1021.977136	2.161716736	0.616511895	3.506366631	0.000454269	0.00977366692263673
hsa-let-7g-3p	0.768717547	-3.385800009	1.925045932	-1.758815181	0.078608905	0.226879361199012
hsa-let-7g-5p	1441.470232	0.585536279	0.695106253	0.842369459	0.399581183	0.56627273405429
hsa-let-7i-3p	8.725517966	-1.495253583	1.45947774	-1.02451277	0.305593142	0.478964968837903
hsa-let-7i-5p	6348.967287	0.878157538	0.377213993	2.328008915	0.01991163	0.112200456483341
hsa-miR-1-3p	7.898716353	-1.018816793	1.099480254	-0.926634916	0.35411612	0.527308383081522
hsa-miR-100-5p	6.601794852	0.885148402	0.845694392	1.046652798	0.295259741	0.469836962264119
hsa-miR-101-2-5p	0.840473684	-3.801529789	1.846313685	-2.058983703	0.0394958	0.158288160408696
hsa-miR-101-3p	3957.299432	0.642788939	1.320080507	0.486931619	0.626306791	0.75114496844425
hsa-miR-101-5p	3.129411899	-1.902032858	1.643779026	-1.157109823	0.247227494	0.42093889868369
hsa-miR-10226	0.00825442	-5.131050313	4.175654434	-1.228801472	0.21914625	NA
hsa-miR-10392-3p	0.057810657	-5.157981001	4.173395634	-1.235919489	0.216488492	NA

Figure 15: Results obtained from Non-BRCA vs. Controls.

Non-BRCA vs. Controls

- The figure above shows the ouput of the result of the comparison of non-BRCA vs Controls. After analysisng the the result, the number of upregulated miRNAs is 6 and the number of downregulated miRNAs is 111.
- Additionally, 56% of the miRNAs are flagged as they have low counts, which is a substantial portion of the total miRNAs analyzed.
- In general, this updated output indicates that a larger number of miRNAs show significant differential expression, particularly downregulation.
- Many miRNAs flagged as having low counts might have been excluded from the significant results.

Non-BRCA vs. Controls

```
out of 1622 with nonzero total read count
adjusted p-value < 0.1
LFC > 0 (up)       : 6, 0.37%
LFC < 0 (down)     : 111, 6.8%
outliers [1]       : 0, 0%
low counts [2]     : 912, 56%
(mean count < 0)
[1] see 'cooksCutoff' argument of ?results
[2] see 'independentFiltering' argument of ?results
```

Figure 16: Summary of results obtained from Non-BRCA vs. Controls.

The Interpretation of the MA plot

- The MA-plot shows that while most miRNAs do not exhibit significant changes in expression (gray points around y=0), there is a notable number of significantly differentially expressed miRNAs (blue points), with a higher prevalence of downregulation compared to upregulation.
- This visualization supports the earlier summary, which indicated a substantial increase in the number of downregulated miRNAs.

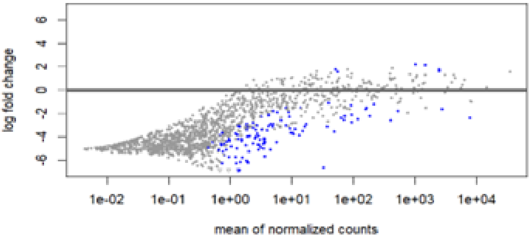

Figure 17: MA plot of Non-BRCA vs. Controls.

Results for Cases (BRCA + non-BRCA) vs Controls

- This output below indicates that the majority of miRNAs with significant differential expression are downregulated.
- A substantial number of miRNAs (78%) have low counts, which may have been excluded from the significant results to focus on more reliable data.
- The number of upregulated miRNAs remains at 1, while the number of downregulated miRNAs is 11.
- This represents fewer significant downregulated miRNAs compared to the previous analysis, which had 111.

Results for Cases (BRCA + non-BRCA) vs Controls

```
out of 1622 with nonzero total read count
adjusted p-value < 0.1
LFC > 0 (up)       : 1, 0.062%
LFC < 0 (down)     : 11, 0.68%
outliers [1]       : 0, 0%
low counts [2]     : 1258, 78%
(mean count < 3)
[1] see 'cooksCutoff' argument of ?results
[2] see 'independentFiltering' argument of ?results
```

Figure 18: MA plot of Non-BRCA vs. Controls.

Interpretation of MA plot of Cases (BRCA + non-BRCA) vs Controls

- The MA-plot shows that most miRNAs do not exhibit significant changes in expression, with a small number of significantly differentially expressed miRNAs being predominantly downregulated.
- This aligns with the statistical summary of 1 upregulated and 11 downregulated miRNAs.
- A substantial 78% of miRNAs are flagged as having low counts, potentially affecting the detection of significant changes.

Interpretation of MA plot of Cases (BRCA + non-BRCA) vs Controls

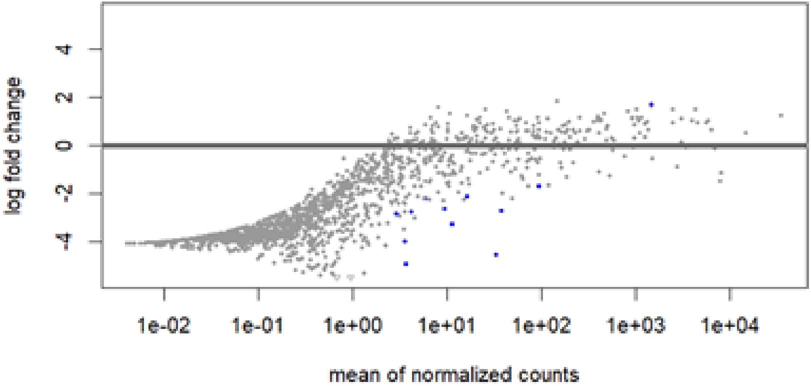

Figure 19: MA plot of Cases (BRCA + non-BRCA) vs Controls.

Visualization of the expression level of miRNAs

- The heatmap visualizes the expression levels of various miRNAs across different conditions, with colour intensity indicating the expression level and clustering showing the relationships between samples and miRNAs.
- This helps in identifying miRNAs that are differentially expressed in BRCA, non-BRCA, and control samples.

Visualization of the expression level of miRNAs and Sample

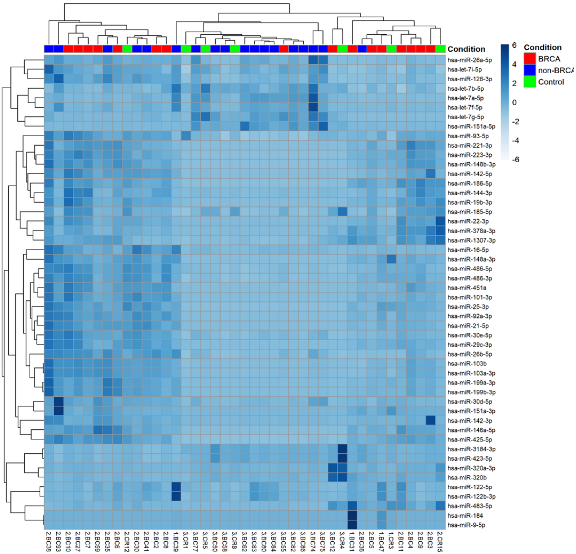

Figure 20: Heatmap of top 50 most variable miRNAs.

Description of the heatmap

- Columns: Each column represents a sample, color-coded by condition:
  - Red: BRCA (Breast Cancer)
  - Blue: non-BRCA (Other conditions related to breast cancer)
  - Green: Control (Healthy samples)
- Rows: Each row represents a specific miRNA, identified by its name (e.g., hsa-miR-142-3p, hsa-miR-16-5p).
- Colour Scale: The colour intensity represents the expression level of miRNAs, with a gradient from blue to white:
  - Dark blue indicates high expression.
  - Light blue to white indicates lower expression levels.

Description of the heatmap

- Dendrograms: The hierarchical clustering dendrograms on the top and left sides show the relationships between samples and miRNAs, respectively:
  - The top dendrogram groups samples based on similarities in their miRNA expression profiles.
  - The left dendrogram groups miRNAs based on their expression patterns across the samples.
- Patterns: The heatmap allows for visual identification of expression patterns:
  - Specific miRNAs may be upregulated or downregulated in different conditions.
  - Clusters of miRNAs that show similar expression patterns can be identified.

Visualization of top 10 highly expressed miRNAs.

```
[1] "Top 10 miRNAs highly expressed in BRCA samples:"
# A tibble: 10 × 2
   miRNA          mean_expression
   <chr>                    <dbl>
 1 hsa-miR-16-5p           26974.
 2 hsa-miR-486-5p          18995.
 3 hsa-miR-320a-3p          9284.
 4 hsa-miR-223-3p           8935.
 5 hsa-miR-451a             8882.
 6 hsa-miR-423-5p           5968.
 7 hsa-miR-3184-3p          5939.
 8 hsa-miR-92a-3p           5546.
 9 hsa-miR-101-3p           5451.
10 hsa-let-7i-5p            5360.
[1] "Top 10 miRNAs highly expressed in non-BRCA samples:"
# A tibble: 10 × 2
   miRNA          mean_expression
   <chr>                    <dbl>
 1 hsa-miR-16-5p           48609.
 2 hsa-miR-486-5p          12772.
 3 hsa-let-7i-5p            8148.
 4 hsa-miR-3184-3p          7588.
 5 hsa-miR-423-5p           7568.
 6 hsa-miR-122-5p           5243.
 7 hsa-let-7a-5p            4400.
 8 hsa-miR-451a             4186.
 9 hsa-let-7b-5p            4155.
10 hsa-miR-92a-3p           4122.
[1] "Top 10 miRNAs highly expressed in Control samples:"
# A tibble: 10 × 2
   miRNA          mean_expression
   <chr>                    <dbl>
 1 hsa-miR-320a-3p         16345.
 2 hsa-miR-16-5p           16129.
 3 hsa-miR-3184-3p         14862.
 4 hsa-miR-423-5p          14372.
 5 hsa-miR-486-5p          10967.
 6 hsa-miR-185-5p           4189.
 7 hsa-let-7i-5p            3983.
 8 hsa-let-7b-5p            3553.
 9 hsa-miR-93-5p            3212.
10 hsa-miR-223-3p           3126.
```

Figure 21: Top 10 miRNAs with the highest mean expression values for each condition.

Conclusion

- The profiling of circulating microRNAs (miRNAs) in familial-hereditary breast cancer (BC) patients has revealed significant differences in miRNA expression related to BRCA status.
- Our analysis identified a few key miRNAs with differential expression between BRCA and control groups, and between BRCA and non-BRCA groups, suggesting their potential as biomarkers for breast cancer diagnosis and prognosis.
- Notably, more miRNAs were downregulated in non-BRCA vs. control comparisons, highlighting the complex regulatory roles of miRNAs in breast cancer. The heatmap visualization further illustrates distinct miRNA expression patterns across different sample groups.
- This study provides a basis for future research to validate these findings and explore the therapeutic potential of these miRNAs in breast cancer treatment.

References

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